

**EXPLORING ENVIRONMENTAL POLICIES TO FOSTER THE
IMPLEMENTATION OF INDUSTRIAL WASTE HEAT RECOVERY IN THE STEEL
MANUFACTURING SECTOR OF INDONESIA: A CASE STUDY BASED ON
SOUTH KOREA EXPERIENCE**

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I. Introduction

As of 2019, Indonesia's greenhouse gas emissions stood at 976.5 Mt CO₂e, ranking 8th globally and first among ASEAN countries (CAIT Climate Data Explorer, 2020). The Indonesian government has set ambitious reduction targets, aiming for a 31.89% reduction (unconditional, CM1) and a 43.2% reduction (conditional, CM2) compared to 2010 emissions by 2030 through an Enhanced Nationally Determined Contribution (NDC) initiative in 2022. Notably, the energy sector is the second-largest contributor to emission reductions after forestry and land sectors. To achieve these goals, the government has integrated climate change considerations into its policy planning processes, implementing measures such as carbon pricing through cap-and-tax schemes, fostering green financing, and promoting a circular economy by repurposing waste as raw materials.

Sector	GHG Emission Level 2010* (Mton CO2-eq)	GHG Emission Level 2030			GHG Emission Reduction				Annual Average Growth BAU (2010-2030)	Average Growth 2000-2012
		Mton CO2-eq			Mton CO2eq		% of Total BaU			
		BaU	CM1	CM2	CM1	CM2	CM1	CM2		
1. Energy*	453.2	1,669	1,311	1,223	358	446	12.5%	15.5%	6.7%	4.50%
2. Waste	88	296	256	253	40	43.5	1.4%	1.5%	6.3%	4.00%
3. IPPU	36	69.6	63	61	7	9	0.2%	0.3%	3.4%	0.10%
4. Agriculture	110.5	119.66	110	108	10	12	0.3%	0.4%	0.4%	1.30%
5. Forestry and Other Land Uses (FOLU)**	647	714	214	-15	500	729	17.4%	25.4%	0.5%	2.70%
TOTAL	1,334	2,869	1,953	1,632	915	1,240	31.89%	43.20%	3.9%	3.20%

Notes: CM1= Counter Measure 1 (*unconditional mitigation scenario*)
CM2= Counter Measure 2 (*conditional mitigation scenario*)

*) Including fugitive.

**) Including emission from estate and timber plantations.

Table 1. Expected BAU and the anticipated decrease in emissions from each sector in Indonesia

Source: *Enhanced NDC Republic of Indonesia, 2022*

The government's primary focus in implementing a circular economy lies in the industrial sector. This sector, apart from serving as a major economic driver, consumes a significant amount of energy and poses substantial environmental challenges, particularly in terms of waste generation, including gases, liquids, and heat. In 2022, the industrial sector, as per IEA report, utilized 37% (equivalent to 166 EJ) of the world's energy, marking an increase from 34% in 2002. This rise in energy consumption can be attributed to the continued expansion of production in energy-intensive industry sub-sectors. Not all the energy consumed is utilized efficiently; in fact, a significant portion is lost. According to Forman et al. (2015), approximately 72% of the world's initial energy consumption is wasted throughout conversion. Additionally, research by Luberti et al. (2021) in industrial sectors revealed that about 20–50% of the energy consumed by the manufacturing sector is also wasted. Heat loss in the industrial and power generation sectors is caused by numerous factors, all resulting from process inefficiencies and culminating in the production of waste heat (Bendig et al., 2013).

Bendig (2013) defined waste heat as essentially heat discharged into the atmosphere, often without consideration for its temperature or potential for reuse. Luberti (2021) further categorizes waste heat based on temperature, distinguishing it into high grade (>600°C), medium grade (240-600°C), and low grade (<240°C), highlighting the significance of this differentiation. Low-grade waste heat is defined by Bendig (2013) as heat that has low economic value and cannot be effectively recovered throughout a process. Additionally, Luberti (2021) suggests that low-grade waste heat remains an underutilized energy source in current practices.

Contrastingly, in the industrial sector where substantial energy is consumed, there is a significant generation of high-grade waste heat. However, only a small fraction of this waste heat has been effectively utilized through industrial waste heat recovery projects. Research conducted by Papapetrou et al. (2018) analyzing the potential for waste heat in each EU manufacturing sector in 2015 (as shown in Fig. 1) revealed a total waste heat potential of 304.13

TWh/year. Among these sectors, Iron and Steel, Non-metallic mineral, and Paper & Printing industries stand out as the top contributors to industrial waste heat. Particularly, the Iron & Steel sector is the primary producer of high-grade waste heat. Hence, the objective of this study is to investigate the utilization of high-grade industrial waste heat in the iron and steel sector.

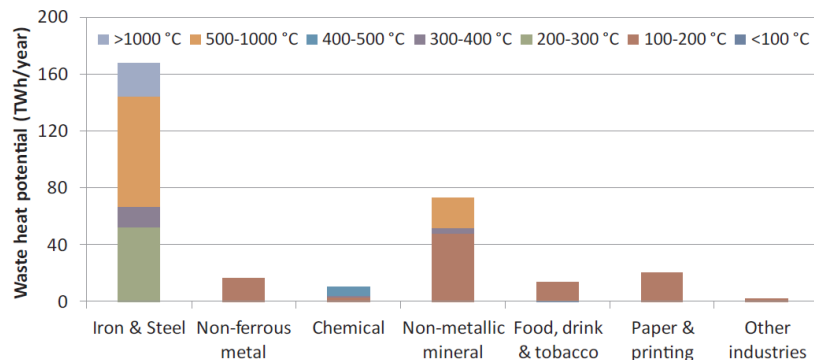


Figure 1. European manufacturing sector waste heat generation across various temperature levels in 2015
Source: Papapetrou et al., 2018

The use of industrial waste heat has enormous potential worldwide; however, in reality, only a small portion of this potential has been utilized. In 2018, research conducted by Luberti (2021) revealed that 2,437.3 TWh of heat was accessible in the EU. Furthermore, Forman (2015) noted that out of the 72% of global energy consumption lost, 63% constitutes a waste heat stream below 100°C, originating from electricity generation, transportation, and industry. The substantial potential is accompanied by significant benefits. Waste heat recovery could decrease the United States' main energy consumption by 12% and carbon emissions by 13%, according to 2011 research by Rattner and Garimella. Furthermore, Kim et al.'s (2018) research on the possibility for synergy between industry and cities, specifically concerning waste heat, highlights an estimated capacity of 49,321 TJ/year for this energy source. It could potentially reduce CO₂ emissions by 243,396 tons annually and reduce fuel costs by 48 million US dollars each year. Considering the potential benefits of waste heat recovery implementation, it is essential to establish an optimal environmental policy to facilitate this process. Thus, **this study aims to** explore the most effective environmental policy for fostering the implementation of industrial waste heat recovery in Indonesia's steel manufacturing sector, drawing insights from the South Korean experience.

II. Literature Review: Policy Instruments and Challenges

Beyond its role as a major economic driver, the industrial sector presents notable environmental challenges, specifically concerning the generation of industrial waste heat. Therefore, harnessing industrial waste heat is pivotal in efforts aimed at energy conservation and emission reduction. Industrial symbiosis, the central idea behind industrial waste heat recovery, industrial symbiosis is defined as a collaborative strategy where one factory's waste serves as another factory's raw material, offering a viable solution to tackle environmental challenges while simultaneously promoting economic growth (Neves et al., 2019). Industrial symbiosis is heavily influenced by existing policies and legislation. The presence or absence of these regulations can either facilitate or hinder industrial symbiosis practices. In cases where policies exist, inappropriate restrictions on waste recovery or usage create barriers to implementation. Conversely, the absence of policies provides no legal basis to guide stakeholders in adopting this concept. The absence of policies prevents industries from pursuing industrial symbiosis, depriving them of the opportunity to derive benefits that can positively affect both corporate performance and the environment (UNIDO, 2022). To overcome these challenges, the government should establish and enforce policies that not only

raise awareness but also incentivize companies to embrace industrial symbiosis practices, as emphasized by Neves et al. (2019).

Park (2013) argues that the government should refrain from strict control over industries regarding the implementation of industrial symbiosis. Instead, it should offer guidance, support, and incentives to encourage industries to voluntarily adopt industrial symbiosis practices. The advancement of industrial symbiosis is driven by the necessity to survive and meet national environmental standards at the most efficient cost. In the case of South Korea, there are no specific regulations governing eco-industrial parks or industrial symbiosis, only a national environmental standard. Park (2013) implies that complex regulations and multiple layers of decision-making processes could impede the smooth implementation of industrial symbiosis initiatives.

In Liu's (2023) assessment of China's potential for recovering heat from industrial waste, challenges in implementing this concept became evident. These challenges stem from the absence of incentive systems encouraging industries to invest in related infrastructure and adopt new technological advancements. Additionally, the lack of financial and tax support policies hampers energy-intensive enterprises from embracing these methods. Furthermore, the absence of government financial backing for scientific and technological innovations within the energy-intensive industry exacerbates the hurdles faced in adopting waste heat recovery initiatives. Liu (2023) suggests that there is potential for large cost savings in the development and maintenance of waste heat recycling facilities by offering policy support and guidance in the form of pricing mechanisms, subsidies, monetary incentives, cost exemptions, and funding and investment options. Additionally, forming public-private partnerships and mobilizing social capital support are essential strategies for proactively contributing to the advancement of waste heat sector initiatives.

Aligned with Liu's, the Industrial Symbiosis concept in Indonesia has faced challenges leading to limited performance improvements. This is primarily due to the absence of regulations and government initiatives supporting its implementation, resulting in insufficient awareness and incentives for industries to adopt the concept. Conversely, Ecosian's (2023) study on the steel industry emphasizes the possibilities of an industrial symbiosis network. By establishing a steam network between waste heat generators and steam users, natural gas-generated steam can be substituted with waste heat-generated steam. This integration not only reduces costs but also lowers greenhouse gas emissions, serving as a successful model that can inspire similar initiatives in the steel manufacturing industries.

III. Methodology

The role of the government, especially through initiatives and regulations, is pivotal in the effective use of industrial waste heat recovery and in overcoming associated challenges. Effective environmental policies need to be developed to incentivize companies to adopt this approach. Kim (2023) categorizes environmental policy instruments into three types: 1) regulatory/administrative (command-and-control); 2) market-based instruments (incentive-based flexible instruments); and 3) voluntary agreements and information strategies. Each instrument offers unique advantages, underscoring the necessity for a comprehensive and tailored policy formulation approach. Instead of relying solely on one instrument, a combination of these instruments into a policy mix is recommended. Jiang (2023) argues that a policy mix involves more than just tool interaction; it necessitates a strategic combination of these instruments. He emphasizes that blending policy instruments from various domains is more effective in enhancing eco-innovation performance compared to relying on a single policy tool. Regarding that, this study aims to address the question: **What is the best environmental policy to foster the implementation of industrial waste heat recovery and overcome the challenges?** This study will briefly outline a case of potential industrial waste heat recovery in

a steel manufacturing sector in Indonesia and its associated benefits. Additionally, a literature review approach will be employed to answer this question by analyzing existing literature and regulations from South Korea's implementation experiences.

IV. Analysis

There exists a potential symbiosis in the steel manufacturing industry, where high-grade waste heat is generated from furnaces. Ecosian's (2023) research conducted in one of the steel companies in Indonesia revealed the possibility of utilizing this high-grade waste heat from the supplier company, converting it into steam that can be utilized by the buyer company in their production processes. At the outset, the supplier's furnace produces high-grade waste heat that is usually discharged into the environment, while the buyer utilizes a Natural Gas-fueled boiler for steam production. A symbiotic opportunity arises between these two companies. The supplier can utilize its excess heat by installing a waste heat recovery boiler, generating steam that can be supplied to the buyer. This not only increases revenue for the supplier but also allows the buyer to reduce costs by utilizing the steam produced from the supplier's waste heat boiler at a lower price than Natural Gas.

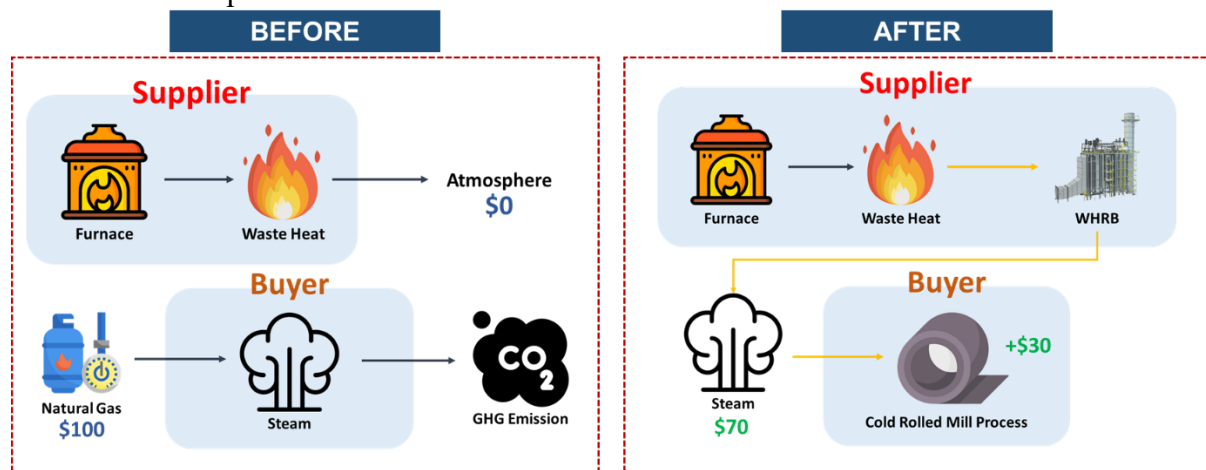


Figure 2. Illustration of the benefit from Supplier and Buyer by utilizing waste heat recovery
Source: Writer interpretation, 2023

Figure 4 illustrates the benefits derived from waste heat recovery for each company, indicating substantial gains for both Supplier in terms of revenue and Buyer in reducing costs and GHG emissions by replacing fossil fuel consumption. Ecosian's (2023) calculations demonstrate positive economic indicators (NPV and IRR), indicating profitability, especially in scenarios with higher natural gas prices. The analysis also reveals a payback period of less than 2 years and a benefit-cost ratio exceeding 2, emphasizing the significant advantages of this symbiotic relationship.

Industrial symbiosis practices in Korea have historically originated within industrial estates, underscoring the pivotal role these estates have played since the early stages of South Korea's rapid industrial growth. These estates have offered substantial economic and social advantages. However, this swift advancement has led to significant environmental deterioration and subsequent challenges to public well-being (Choi, 2012). Due to the harm that their activities have caused, these estates have been the subject of conflicts and increased public attention. With an emphasis on industrial symbiosis, the National Eco-Industrial Park (EIP) project was started by the Korean government in 2003 through the Korea National Cleaner Production Center (KNCPC) to solve these problems. This program attempted to strike a balance in accordance with the Ministry of Trade, Industry, and Economy's (MOTIE) initiatives to support progressive manufacturing development with an emphasis on environmental

sustainability. Regrettably, the top-down strategy that led to the private sector's initial lack of engagement over the initial two years produced few significant results (Kim, 2017).

The Korean government adopted a more business-oriented approach by entrusting the role of implementation to the Korea Industrial Complex Corporation (KICOX). With this new strategy, business involvement increased notably in the program's third year, leading to a surge in projects. According to Kim (2017), by 2015, KICOX had received 595 project proposals, with 388 projects receiving funding for further study and improvement, and 197 projects being implemented. The program had a major impact since, between 2005 and 2014, initiatives using cleaner manufacturing methods and Industrial Symbiosis facilities together reduce 1.09 million tons of other hazardous gas emissions and 6.48 million tons of CO₂ emissions. Furthermore, allocations for research and advancement, along with investments in industrial infrastructure, generated 848 jobs, facilitated technology advancements, and resulted in the creation of 56 new patents with 100 awaiting. As of 2015, participating companies collectively benefited from substantial savings approximately USD 1,680 million, achieved through resource conservation, waste sales, and by-product transactions facilitated by the Industrial Symbiosis systems.

In my analysis of the factors contributing to the success of the KICOX program in promoting industrial symbiosis in Korea, I found support for my hypothesis. This analysis indicates that a regulatory (command and control) approach is less effective in encouraging industries to implement industrial symbiosis systems compared to a market-based approach involving incentives and various schemes. The following findings demonstrate the effectiveness of this program:

- a. **The bottom-up approach:** concentrates on involving and enticing companies by leveraging the expertise of local professionals and their networks. This strategy aims to increase industry and community involvement in the initiative.
- b. **The decentralization strategy:** focuses on community building by establishing regional EIP centers and appointing local coordinators, which aim to create stakeholder communities based on shared interests and enhancing business involvement in the program.
- c. **Success spillover information:** centered on producing immediate results, proving financial viability, and demonstrating other industries how to execute successful strategies.
- d. **Encouraging participation through incentives:** the Korean government implemented a successful funding scheme that encouraged private involvement and investment. The government provided financial support for the research and development requirements of selected projects. Funding allocations were based on factors such as the recipient company's size, prospective effect, level of technological innovation, and other particular criteria.
- e. **Waste information sharing:** willingness to open the waste disclosure information and identification of the waste also types of company in the industrial estates makes it easier to seek the potential symbiosis between companies.

Korea Government legalize the market-based approach environmental policy by enacting Act Number 14839 of 2017 about Act on The Promotion of The Conversion into Environment-Friendly Industrial Structure. The main point of this act are: 1) The MOTIE should develop and execute policies to promote the transition to a resource-circulating industrial structure in consultation with the Minister of Environment, fostering collaboration among Ministries; and 2) the government will provide the required funds for executing the specified projects or offer essential support, which may include technology development, establishing a comprehensive management system, expert training, and collaboration with local communities. This Act supported by Presidential Decree Number 32586 of 2022 about Enforcement Decree of The Act on The Promotion of The Conversion into Environment-Friendly Industrial Structure, including providing information or data for network connecting between companies, publicity of successful cases, feasibility studies, and support for obtaining certification of GHG reduction.

Nevertheless, there are still regulations that can hinder the implementation of industrial symbiosis practices. Choi (2012) highlighted challenges arising from several regulations, especially strict waste recycling regulations, which include: 1) The Waste Control Act, which focuses on safe waste treatment rather than recycling or exchanging of waste. This Act does not differentiate between hazardous wastes that must be disposed of and potential materials that could be incorporated into industrial processes; and 2) The Act on the Promotion of Saving and Recycling of Resources, which uses primary components of wastes as criteria to determine the legality of recycling. This criterion has posed obstacles in the environmental industries. Neves (2019) emphasizes that the lack of appropriate existing policies, such as restrictions on wastewater usage, low taxes on landfill disposal, and inadequate funding to promote this practice, creates difficulties in the process of industrial symbiosis implementation. To address this issue, the government should implement policies that raise awareness and incentivize companies to adopt industrial symbiosis practices.

V. Conclusion and Recommendation

The case of industrial waste heat recovery in the steel manufacturing sector in Indonesia demonstrates significant potential benefits, including: 1) Economic advantages through enhanced cost competitiveness, resulting in annual profits and cost savings; 2) Social benefits by increasing customer satisfaction through the implementation of cleaner production practices; and 3) Environmental benefits achieved by recycling waste heat and reducing the consumption of fossil fuels, thereby mitigating greenhouse gas emissions.

Drawing insights from the benefits observed and lessons learned from South Korea's experience, the promotion of industrial waste heat recovery in the steel manufacturing sector in Indonesia necessitates the adoption of market-based environmental policies and voluntary information-based approaches. The South Korean case highlights the limitations of top-down regulations (command and control-based) in generating significant impact. This market-oriented policy approach should be coupled with stringent environmental standards. To meet these high standards, the government can incentivize companies through funding initiatives, encouraging the efficient implementation of waste heat recovery projects. The government can provide several recommended incentives, including:

1. **Institutional Reform:** Establishing specialized institutions or management systems to gather waste information and identify potential symbiotic opportunities. This could involve creating regional waste heat recovery centers.
2. **Financial Support:** Offering subsidies, tax benefits, research and development (R&D) funding, support for new technological advancements, investment and financing assistance.
3. **Services:** Providing expert training, fostering collaboration with local communities, conducting feasibility studies, assisting in obtaining certifications for GHG reduction, promoting investments, and recognizing green companies through awards.

The government can implement specific voluntary information-based policies, which include:

1. **Publication of Successful Implementations:** Showcasing the benefits of waste heat recovery projects, such as cost reduction and greenhouse gas (GHG) emission reduction, to inspire other companies to participate. This is especially crucial for addressing current issues like climate change.
2. **Raising Social Awareness:** Promoting public-private partnerships and social capital investment in the industrial sector to actively participate in waste heat industry project development.

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